## OS1-1

# Can you hear the pain in a baby's cry? Unlocking the secrets of experience-driven perception

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This study investigates the role of experience in detecting pain in infant cries, challenging the assumption that universal acoustic features enable innate recognition. Through psychoacoustic experiments, we demonstrate that adults without caregiving experience fail to distinguish pain cries (e.g., post-vaccination) from discomfort cries (e.g., during bathing), even when familiar with the infant. Conversely, parents and professional caregivers accurately identify pain cries, including from unfamiliar infants, with parents of young children excelling across contexts. Acoustic analyses reveal pain cries are longer, louder, and marked by nonlinear phenomena (e.g., chaos, sub-harmonics), which overpower individual identity cues to prioritize distress signaling. Machine learning confirms identity remains encoded but is significantly weakened in pain cries, reflecting an evolutionary trade-off favoring survival over recognition. We further explore intergenerational decoding, finding that older adults with caregiving experience match or exceed younger adults in identifying pain cries. This supports the cooperative breeding hypothesis, highlighting grandparents' role in alloparental care. Playback experiments with natural and synthetic cries show nonlinear phenomena, especially chaos, drive adult pain perception. These findings underscore that while universal acoustic traits underpin infant cries, experiential learning shapes their interpretation, balancing static identity and dynamic distress cues. This research bridges auditory cognition, evolutionary biology, and intergenerational caregiving, offering implications for infant care and communication development.

## OS1-2

# Insectivorous bats supplement echolocation with vision to hunt insects

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Animals can improve decision-making by integrating information from multiple senses, which is especially advantageous when living in fluctuating environments. Understanding how wild predators may use multimodal sensing when hunting prey in split-second interactions remains largely unexplored. As nocturnal hunters, bats rely on echolocation to orient themselves in space and catch evasive prey, yet they have remained functional vision. We therefore investigated whether bats increase sensory redundancy during orientation and prey capture by complementing echolocation with visual cues. We equipped wild common noctule bats with high-resolution sound and movement loggers to measure their echolocation behaviour when commuting and foraging in dark and lit environments. While commuting, the bats maintained consistent echolocation sampling across light levels. However, when tracking prey in illuminated environments, bats emitted calls less frequently and with slower decreases in call levels compared with dark conditions. This indicates that hunting bats supplement acoustic with visual information in illuminated environments. This bimodal sensing strategy likely explains the faster flight approaches to prey in lit conditions, suggesting a more efficient hunting strategy when integrating information from two rather than a single sensory modality. Even with ample visual cues (>100 lux), bats continued to echolocate suggesting that biosonar remains the primary sensory modality. Our findings demonstrate how a wild insect predator uses multimodal sensing by regulating its primary sensory modality based on the availability of its secondary sensory modality for efficient hunting in highly dynamic and complex environments.

## OS1-3

# Acoustic differences between sister species birds and the impact on interspecific recognition

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Acoustic characteristics play a crucial role in species delimitation among birds. Descriptive approaches—using comparisons of temporal and frequency features in vocalizations as evidence for taxonomic revisions—are widely employed in taxonomic studies. However, whether the manually selected acoustic features align with birds' perceptual recognition remains to be verified through playback experiments. Locustella mandelli (Russet Bush Warbler) and Locustella chengi (Sichuan Bush Warbler) are two morphologically similar sister species whose songs exhibit distinct differentiation. The song frequency of L. mandelli (~5 kHz) is significantly higher than that of L. chengi (~3 kHz), and the duration of high-energy segments also shows differences. This study designed four playback stimuli to test the responses of these two species to different frequency and temporal features: a. Normal L. mandelli song; b. Frequency-decreased L. mandelli song; c. Normal L. chengi song; d. Frequency-increased L. chengi song. Results showed that L. mandelli responded strongly only to stimuli a and d, whereas L. chengi responded primarily to stimuli b and c. These findings indicate that interspecific recognition between the two species relies mainly on frequency rather than temporal characteristics. This study highlights the necessity of incorporating playback experiments in taxonomic research to assess the role of acoustic divergence in the speciation of closely related bird species.

## OS1-4

# The sounds of pleasure and disgust in human nonverbal vocalizations

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Beyond language, the human voice conveys rich information, notably through nonverbal vocalizations like laughter and cries, reflecting both static traits and dynamic emotional states. Sensory experiences strongly influence vocal production. However, little is known about how external stimuli, like odors, shape vocal responses. As Darwin suggested, such expressions may have deep evolutionary roots, akin to animal food-related calls signaling edibility. However, unlike most other mammals, humans can voluntarily modulate such nonverbal vocalizations with ostensible adaptive benefits—for instance, reinforcing food learning in parent-child interactions or facilitating social bonding. The efficacy of such vocal control remains unclear.
To fill this gap, we recorded human participants producing nonverbal vocalizations in response to pleasant and unpleasant odors. We then tested their ability to exaggerate, suppress, or fake their vocal responses—for example, expressing pleasure for a disgusting smell. Acoustic analyses examined how odor preferences are encoded in the voice and which vocal parameters can be voluntarily controlled. Finally, playback experiments assessed whether listeners can reliably judge pleasantness and disgust from the voice alone, while detecting inauthentic responses.
We hypothesize that the acoustic forms of these nonverbal vocal responses should reflect their functions. Crucially, vocalizers may be able to exploit voluntary vocal control to manipulate these vocalizations and influence social interactions to their benefit, while listeners would benefit from detecting deceptive social cues. Our results provide new insights into the interplay between sensory experience, vocal production, and its control in humans, emphasizing an evolutionary arms race between signallers and receivers in deceptive vocal signalling.

## OS1-5

# Aquatic percussions: sound production mechanism in the Cichlid family

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In Teleostean fish, sound-producing mechanisms described in the literature typically involve the contraction of very fast muscles, the sonic muscles. They compress a gas-filled pouch organ, the swim bladder, at a high specific rate resulting in sounds. Well... the Cichlid family, although known to be highly vocal, does not possess sonic muscles. In this study, we show that species of this family have developed a very unique sound-producing mechanism compared to other Teleostean fish. The swim bladder and the skeleton form an inseparable system: the ventral part of the swim bladder is stretched by the ribs. To produce sound, cichlids compress the ribs associated with their swim bladder, and relaxation allows the membrane to resonate.

## OS1-6

# Sub-surface behaviors of Hector's dolphins increase risk of bycatch

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Hector's dolphins, endemic to New Zealand, suffer from bycatch in fishing gear. Fishing restrictions in inshore areas have reduced bycatch, although dolphins have been recorded outside these areas. We studied the sub-surface movements and foraging behavior of 11 Hector's dolphins using high-resolution biologging tags to gain insight into their bycatch risk. Tagged dolphins travelled up to 15 km offshore, well beyond the boundaries of fishery exclusion zones, and regularly dove to the seafloor to forage at depths of up to 122 m. During these dives, the dolphins exhibited specific body postures that could increase their risk of bycatch. Additionally, some dolphins produced very few echolocation clicks while travelling close to the seafloor, potentially limiting their ability to detect nets. These sub-surface behaviours may make Hector's dolphins more vulnerable to bycatch than previously thought, highlighting the need to reassess the efficacy of current protection measures.

## OS1-7

# Does acoustic camouflage of a moth's (*Lepidoptera*) body scales trade-off with thermal insulation?

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Multiple properties derived from the same structure often co-vary with important implications for an organisms' survival or reproduction. Functional conflicts often arise due to shared mechanisms, but such trade-offs may only become apparent in populations at the extremes of trait variation. We investigated the potential trade-off between two properties of the thoracic and abdominal scales in adult moths (Lepidoptera): thermal insulation and sound absorption. These hair-like scales retain heat generated by the vibration of wing muscles located in the thorax, but they also acoustically camouflage the moth, reducing the energy reflected to a predating echolocating bat compared to a bare cuticle. Previous research has suggested two mechanisms for acoustic camouflage: porous absorption or resonant absorption. Porous absorption dissipates sound energy in trapped air pathways, a material feature that also reduces heat transfer. Meanwhile, resonant absorption involves energy loss through vibration of a springloaded structure (i.e. a scale), this mechanism of absorption has minimal implications on the thermal properties of the material. In this study we quantified both traits across moth species adapted to a range of microclimatic temperatures and predation pressures. We quantified the thermal insulation capabilities of the scale through an internal heating experiment and measured target strength difference of intact and descaled moths using a custom-built robotic bat head. Additionally, we investigated the resonant absorption properties of the body scales using laser Doppler vibrometry. By analyzing correlations between thermal and acoustic trait values and underlying mechanisms, we provide new insights into the multifunctionality constraints on moth scales.

## OS1-8

# Bioacoustics in the era of open access to standardized data

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Bioacousticians have existed as long as humans, with many cultures passing down species identifications and imitations of calls that modify wildlife behavior. Modern bioacoustics started with the development of electronic audio recorders and analyzers such as sound spectrographs. Individual labs tended to develop their own categorizations and names for call categories, and seldom shared data, which led to confusion and conflict. As pioneers in the field have retired, we risk losing data whose age is particularly valuable for some applications. Here I discuss the development of a community of ocean acousticians, narrowly marine from a bioacoustics perspective, but much broader in terms of scientific applications, who are implementing a plan to coordinate recordings globally, to standardize data and analyses, and to develop open-access databases of curated ocean sound data. Ocean Sound has been designated as an essential ocean variable by the Global Ocean Observing System. Derived variables that estimate wind, rain, and sea ice are important for assessing climate change, variables that detect earthquakes and anthropogenic noise are important for managing threats such as tsunamis and noise, and variables that detect physical, anthropogenic, and biotic sources of sound are important for defining soundscapes. As bioacousticians specialize in data collection, calibration and QA/QC, and increasingly varied and sophisticated analyses, the establishment of large, stable, openaccess databases that provide scientific credit to the collector and availability over decades for mul(ple analyses will be critical for the future of our field. Challenges include stability of funding for long-term recording stations and databases.

## OS1-9

# Foraging strategies of *Micronycteris microtis* hunting in the understorey

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The neotropical gleaning bat Micronycteris microtis has been shown to find silently resting prey on vegetation with echolocation. Previous experiments indicate that they approach prey on leaves from oblique angles allowing them to use acoustic specular reflections off leaves to uncover prey echoes. However, it is not clear which acoustic or behavioural strategies bats follow to efficiently inspect the vast number of leaves in an acoustically highly complex environment, like a forest understorey. Given that active flight is energetically costly and the great majority of leaves in the forest would not hold resting prey; bats should use an efficient selection mechanism that focuses the bat's attention on leaves with a high probability of the presence of prey. To study the acoustic and behavioural mechanism underlying the foraging strategy developed by M. microtis we performed a behavioural experiment in a dark flight cage containing an artificially created understorey vegetation. This artificial vegetation consisted of 8 custom-made 'plants' each with three leaves placed at different heights and in differently oriented planes. Recordings from 4 infrared-cameras synchronised with a 24-ultrasonic microphone array (one microphone/leaf) were used to study 15 bats, each individually observed for 2.5-hours while searching for prey randomly placed on 8 of the 24 leaves. Here we will present preliminary results from a semi-supervised analysis of the bats' flightpaths and the corresponding echolocation behaviour.